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Dissecting Sequence, Structure, and Data Recency Effects in Enzyme Commission Prediction under Temporal Distribution Shift

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Abstract

Background: Enzyme Commission (EC) prediction is increasingly evaluated with large protein language models and structure-derived features, but headline benchmark scores often conflate model architecture, training-set recency, sequence overlap, label-vocabulary coverage, and temporal distribution shift. This makes it difficult to determine whether a new representation improves biological generalisation or mainly benefits from benchmark design.

Results: We analyze Contact-EC, a gated additive ESM-2/contact-map fusion model, as a controlled probe for decomposing temporal EC prediction performance. On a Swiss-Prot 2023-01 temporal holdout ($N = 124$, three seeds), fusion reaches Level-4 micro F1 of 0.6241 ± 0.0170, outperforming ESM-2-only and contact-only baselines; topology perturbations indicate that the contact-map encoder uses structural information beyond simple map-density cues. HIT-EC leads on this one-year horizon (0.8471 vs. 0.6241), indicating that structure-aware fusion alone does not close the short-horizon temporal gap. A larger

SP-2024 temporal evaluation ($N = 1{,}226$, three seeds) reveals a reversal under our mapped evaluation: HIT-EC falls to 0.4578 (-38.9 pp) while Contact-EC improves to 0.6819 ± 0.0026 ($+5.8$ pp), yielding a $+22.4$ pp advantage; a vocabulary-stratified analysis indicates that most of HIT-EC's decrease is not explained by label-vocabulary mismatch. A Foldseek/TM-score-disjoint split shows that fusion retains the highest micro F1 under explicit fold-level shift.

Conclusions: Contact-EC is best interpreted as a reproducible benchmark-decomposition instrument rather than a replacement for curated homology transfer. The results show that EC prediction benchmarks should explicitly report temporal cutoff, sequence and fold relatedness, label-vocabulary coverage, data recency, and whether the evaluation is closed-set or label-shifted.

Key words enzyme commission number; contact map; protein language model; temporal evaluation; sequence–structure fusion; out-of-distribution generalisation

1. Background

Enzyme Commission (EC) numbers [1] provide a four-level hierarchy for classifying enzyme-catalysed reactions, from broad reaction class to substrate-specific function. Accurate EC assignment supports metabolic reconstruction, pathway annotation, enzyme engineering, and drug-target discovery, but experimental characterisation remains slow relative to the growth of protein sequence databases.

Computational EC prediction has progressed from alignment-based transfer [2] and 1D convolutional classifiers on one-hot sequences [3, 4] to large protein language model (PLM) representations. ESM-2 [5] encodes residue-level evolutionary and structural information from large sequence corpora, while systems such as HIT-EC [6], CLEAN [7], EnzBert [8], and ECPick [9] apply hierarchical, contrastive, or evidential heads to PLM embeddings. Structure-aware approaches derive contact-map features from predicted structures [10, 11]: DeepFRI [12] uses graph convolutional networks over residue contact graphs, and CLEAN-Contact [13] integrates AlphaFold-derived maps into a contrastive learning framework. ProteInfer [14] further extends function prediction with dilated convolutional sequence profiles. However, reported scores are often obtained on same-distribution or near-contemporaneous splits. This creates a reliability problem for biological deployment: a model may rank highly because it has access to more recent annotations, an easier closed-set label vocabulary, or near-duplicate sequences, rather than because it generalises to newly annotated enzymes.

We study these factors using Contact-EC, a gated sequence–structure fusion model combining ESM-2 650M embeddings with ResNet-50 contact-map features. The goal is not to claim a new state-of-the-art EC predictor: HIT-EC remains stronger on the shorter SP-2023-01 temporal holdout. Instead, Contact-EC is used as an instrument for benchmark decomposition. We ask three questions that are central for journal-scale model assessment. First, how different are same-distribution, sequence-disjoint, and temporal EC evaluations? Second, do contact maps contribute signal complementary to ESM-2 after controlling for topology-destroying perturbations? Third, how much of temporal performance is explained by data recency, closed-set label coverage, and fine-tuning strategy rather than by architecture alone?

Our analysis makes seven contributions. We provide a temporal label-shift audit for Swiss-Prot 2023-01, separating 124 complete known-EC proteins from 344 partial or novel-label cases. We show that contact maps alone match ESM-2 on hard sequence-disjoint validation, while fusion improves temporal micro F1 over either single modality. We test whether the contact branch uses fold topology rather than contact-map density through shuffled, diagonal-only, and density-matched controls. We add a Foldseek/TM-score-disjoint split to test structural generalisation beyond sequence-disjoint evaluation. We extend temporal evaluation to SP-2024 ($N = 1{,}226$), observing a HIT-EC reversal under the mapped evaluation: HIT-EC falls -38.9 pp while Contact-EC improves $+5.8$ pp, and is higher than HIT-EC by $+22.4$ pp on the longer horizon, widening the modality gap from 17 pp to 29 pp. We decompose data recency and fine-tuning, finding that newer Swiss-Prot data helps whereas end-to-end fine-tuning over-adapts. Finally, we translate these results into reporting requirements for EC prediction studies: temporal cutoff, sequence overlap, fold-level relatedness, label-vocabulary coverage, partial-label frequency, structure availability, and whether each result is measured on a closed-set or label-shifted benchmark.

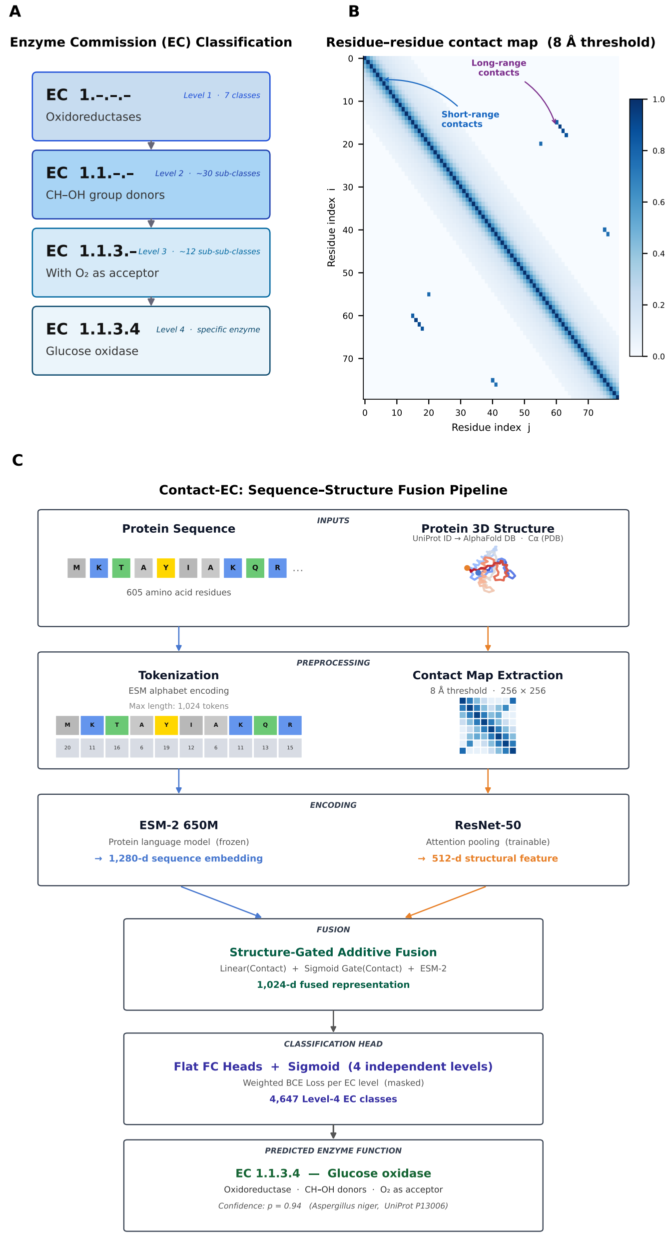

Figure 1 Contact-EC overview. **(A)** EC number four-level hierarchy illustrated with glucose oxidase (EC 1.1.3.4). **(B)** Example protein contact map (8 Å threshold); short-range backbone contacts form the diagonal band; long-range contacts (blue arrows) encode tertiary fold topology. **(C)** Contact-EC pipeline: ESM-2 (1280-d) and a ResNet-50 contact-map encoder with attention pooling (512-d) are combined by structure-gated additive fusion into a 1,024-d fused representation, then a flat FC + sigmoid head predicts all 4,647 Level-4 EC-Bench classes simultaneously (Contact-EC-Hier replaces this head with a two-layer Transformer over four EC-level tokens; Section 2.2).

2. Methods

2.1. Data and Splits

The primary controlled experiments follow EC-Bench: Swiss-Prot February 2018 for training (201,865 proteins after local preprocessing) and Swiss-Prot 2023-01 for temporal testing. Sequences were truncated to 1,024 residues. AlphaFold structures were retrieved from the AlphaFold Protein Structure Database [11] when available; proteins without structures received zero contact maps. Contact maps were built from Cα distances using an 8 Åthreshold, resized to 256×256, and encoded in three channels: all contacts, short-range contacts ($|i - j| < 12$), and long-range contacts ($|i - j| \geq 12$).

We distinguish four evaluation regimes. A random 80/10/10 Swiss-Prot split measures in-distribution performance. A MMSeqs2 [15] 30% identity cluster split measures sequence-disjoint generalisation. An EC-Bench-style hard validation set (`val_hard`; 6,247 proteins, ≤30% sequence identity) provides a shared validation protocol rather than an independent hidden test. Of 468 Swiss-Prot 2023-01 temporal proteins, only 124 have complete Level-4 EC labels inside the SP-2018 vocabulary; the remaining 344 contain partial or novel labels. A final temporal audit found no accession or exact-sequence overlap between these 124 proteins and either SP-2018 or ExpA training, but temporal evaluation alone does not guarantee fold-disjointness.

To directly evaluate fold-level shift, we constructed an additional Foldseek-disjoint split from the SP-2018 training corpus. Proteins with available AlphaFold structures were clustered by Foldseek [16] using TM-score threshold 0.50, coverage 0.80, alignment type 1, cluster mode 1, and sensitivity 7.5. Cluster assignment produced 170,573 training, 25,565 validation, and 16,725 test proteins across 3,051/381/381 structure clusters, with zero protein-ID overlap across partitions. This split is used only for the fold-disjoint audit and all models are retrained on the corresponding training partition.

2.2. Model

The sequence branch uses ESM-2 650M [5] (`facebook/esm2_t33_650M_UR50D`) and extracts the 1280-dimensional `[CLS]` embedding. The structure branch applies a ResNet-50 encoder [17] with attention pooling to the three-channel contact map and projects the output to 512 dimensions. The two representations are projected to a shared 1024-dimensional space and combined by structure-gated additive fusion:

$$\tilde{\mathbf{e}} = \phi(W_e \mathbf{e}), \qquad \tilde{\mathbf{s}} = W_s \mathbf{s},$$

$$\mathbf{g} = \sigma(W_g \mathbf{s}), \qquad \mathbf{f} = \tilde{\mathbf{e}} + \mathbf{g} \odot W_a \tilde{\mathbf{s}}.$$

The gate $\mathbf{g}$ is a function of the raw structure feature $\mathbf{s}$ alone and is not conditioned on the sequence branch; W_a is implemented as a single-head attention layer applied to a single structure token, which is mathematically equivalent to a learned linear map, since softmax over one key is identically 1 regardless of the query. This lets the gate learn to suppress an uninformative or missing contact map without needing sequence context to detect it, giving fallback behaviour to ESM-2-only performance. Contact-EC-Hier passes the fused vector through a two-layer Transformer over four EC-level tokens; the flat variant uses fully connected sigmoid outputs. To audit whether this learned fusion block is necessary, we also train three same-encoder flat-head controls: concatenation followed by an MLP, element-wise summation, and a feature-wise gated MLP (Supplementary architecture-baseline table).

2.3. Training and Evaluation

Models are trained with weighted binary cross-entropy across EC levels, with weights $(0.1, 0.1, 0.2, 0.6)$ for Levels 1–4 and masks for partial annotations. Phase 1 trains downstream modules for 30 epochs with frozen ESM-2; Phase 2 partially unfreezes the final two ESM-2 blocks for 20 epochs at a lower learning rate for Contact-EC-Hier. AdamW [18] with cosine annealing is used throughout, and checkpoints are selected only on validation data. The main training recipe uses batch size 64, Phase-1 learning rate 10^{-3}, Phase-2 learning rate 10^{-4}, weight decay 5×10^{-4}, dropout 0.3–0.4 depending on the split, 8 Åcontact thresholds, 256×256 contact maps, and seeds 42/43/44 for repeated temporal and Foldseek-disjoint comparisons (Supplementary training hyper-parameter table). Level-4 micro F1 is the primary metric, with macro F1, weighted F1, rare-EC F1, precision, and recall reported for interpretation. Hyper-parameters were fixed before temporal testing.

2.4. AI-assisted tools

AI tools were used for grammar checking, LaTeX formatting assistance, and code refactoring during manuscript preparation. All scientific content, experimental design, analysis, and interpretation were performed by the author.

3. Results

3.1. EC-Bench-Style Validation

On EC-Bench `val_hard`, the checkpoint-selected hard validation protocol, contact maps alone reach nearly identical Level-4 micro F1 to ESM-2 650M (0.7650 vs. 0.7655; Table 1). This parity suggests that fold topology captures substantial EC-discriminative information even without sequence embeddings. Fusion improves substantially: Contact-EC reaches 0.8879, while Contact-EC-Hier reaches 0.8698. Training dynamics in Supplementary Fig. S1 show that B3 converges within about 10 epochs and that fusion continues to improve during partial unfreezing.

Table 1 EC-Bench hard validation results (`val_hard`, 6,247 proteins, ≤30% identity), used for validation/checkpoint selection.

Model	L4 Micro F1	L4 Macro F1	Best Epoch
B1 (ESM-2, flat FC)	0.7655	0.1353	30
B2 (ESM-2, hier. FC)	0.4204	0.0383	30
B3 (Contact map only)	0.7650	0.1335	10
Contact-EC (flat FC)	**0.8879**	**0.2294**	29
Contact-EC-Hier	0.8698	0.2141	20 (Phase 2)

3.2. Temporal Label Shift

The official Swiss-Prot 2023-01 temporal set contains 468 proteins, but only 124 are complete known-EC cases that can be scored within the SP-2018 closed-set label space (Table 2). The remaining 344 proteins expose an important benchmark limitation: Level-4 closed-set scoring is undefined for incomplete partial EC labels and cannot assign novel Level-4 classes outside the training vocabulary. We therefore report main temporal comparisons on the 124 evaluable proteins and use the full 468 proteins as a label-shift audit. This separation is essential for interpreting temporal EC prediction: a low score on all 468 proteins may reflect open-vocabulary label shift rather than a failure to rank known EC classes, whereas a high score on the 124-protein subset may overstate deployability on newly annotated enzymes. Supplementary hierarchical-prefix evaluation further shows that partial annotations retain useful lower-level supervision and should not be counted as Level-4 failures.

Table 2 Swiss-Prot 2023-01 label-shift audit. Partial annotations lack complete Level-4 ground truth and are not assigned Level-4 F1.

Subset	N	Interpretation	Closed-set L4 status
Complete known EC labels	124	Main evaluable subset	Scored at L4
Partial EC annotations	309	Incomplete Level-4 labels	Not scored at L4
Novel EC labels	35	Outside SP-2018 vocabulary	Out of vocabulary
All temporal proteins	468	Label-shift audit	Coverage audit

3.3. Temporal Performance

On the complete temporal subset, sequence–structure fusion outperforms both single-modality baselines (Table 3). Across three seeds, Contact-EC improves micro F1 from 0.4508±0.0203 for B1 and 0.4244±0.0207 for B3 to 0.6241±0.0170. The 3B flat-fusion variant reaches 0.6316 in a single run, a modest gain over the 650M fusion mean relative to the 4.5× parameter increase. Contact-EC-Hier improves over single-modality baselines but underperforms the flat head, mainly through lower recall. HIT-EC remains higher at 0.8471, so we interpret Contact-EC as a decomposition model rather than a replacement for stronger sequence-only systems.

Case-wise and baseline audits support this framing. Among 124 temporal proteins, HIT-EC and Contact-EC both recover at least one correct EC for 63 proteins, HIT-EC alone recovers 45, Contact-EC alone recovers 2, and both miss 14. Changing the global threshold only modestly changes Contact-EC micro F1 (0.6032 at 0.5 versus 0.6167 at 0.65), and MMseqs2 top-hit EC transfer reaches 0.5852. Trainable fusion controls remain below Contact-EC: concatenation reaches 0.5922±0.0150 micro F1, summation 0.5505±0.0230, and a gated MLP 0.5543±0.0525 (Supplementary architecture-baseline table).

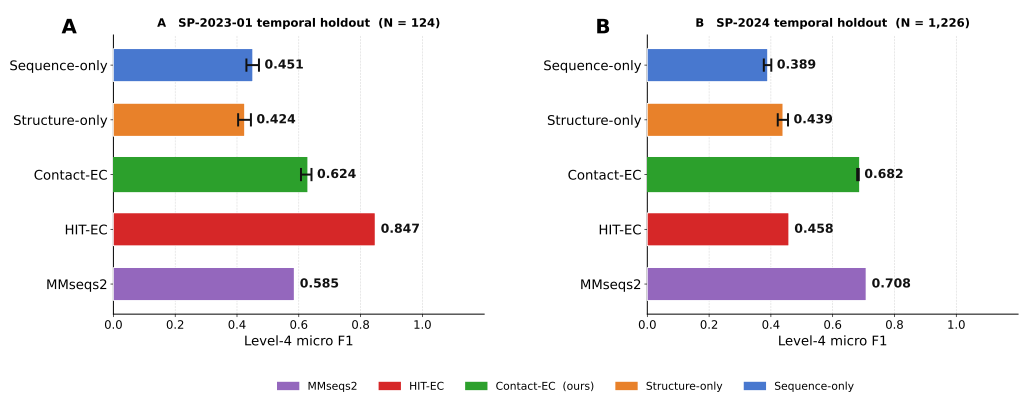

Figure 2 Level-4 micro F1 on two temporal holdouts. **(A)** SP-2023-01 (N=124). **(B)** SP-2024 (N=1,226). Error bars: ±1 s.d. across three training seeds (42/43/44); HIT-EC and MMseqs2 are single deterministic runs. Contact-EC is the only neural model that improves from Panel A to Panel B (+5.8 pp), while HIT-EC regresses −38.9 pp.

Table 3 Temporal and external benchmark results. Swiss-Prot 2023-01 uses the complete known-EC subset ($N = 124$). B1, B3, and Contact-EC flat FC temporal values are mean $\pm$ sample standard deviation across three seeds when shown; other rows are single-run or externally reported. Bold denotes the best value among our SP-2018-trained models. †ExpA is evaluated on the Level-4 encoder-evaluable intersection subset ($N = 99$); Contact-EC trained on SP-2018 reaches 0.6467 ± 0.0210 micro F1 on the same subset.

	Swiss-Prot 2023-01			Price-149
Model	Micro F1	Weighted F1	Rare-EC F1	Micro F1
HIT-EC (own split)	0.930	–	–	–
HIT-EC (this test)	0.8471	0.8134	0.8322	–
CLEAN-Contact	–	–	–	0.525
B1 (ESM-2, flat FC)	0.4508 ± 0.0203	0.3387 ± 0.0194	0.3292 ± 0.0272	0.0324
B2 (ESM-2, hier. FC)	0.1111	0.0835	0.0000	0.0000
B3 (Contact map only)	0.4244 ± 0.0207	0.3325 ± 0.0237	0.3354 ± 0.0254	0.0000
Contact-EC (flat FC)	0.6241 ± 0.0170	0.5278 ± 0.0222	$\mathbf{0.5257 \pm 0.0313}$	0.2500
Contact-EC-Hier	0.5690	0.4528	0.3621	0.1263
Contact-EC-3B (flat FC)	**0.6316**	**0.5477**	0.4786	**0.2915**
Contact-EC-ExpA (recent data)†	0.7417 ± 0.0182	0.6768 ± 0.0168	0.5474 ± 0.0678	0.2764
Contact-EC-E2E (fine-tuned)	0.6703	0.6382	0.5091	0.2522

Bootstrap confidence intervals (1,000 resamples) confirm the main differences: Contact-EC-3B [0.547, 0.711], Contact-EC [0.511, 0.686], Contact-EC-Hier [0.476, 0.664], B1 [0.315, 0.521], and B3 [0.254, 0.473]. McNemar tests further support Contact-EC over B1 ($\chi^2 = 22.4$, $p < 0.001$) and over B3 ($\chi^2 = 37.0$, $p < 0.001$). Supplementary late-fusion diagnostics show that simple post-hoc averaging or maximum pooling of B1 and B3 probabilities does not reproduce the learned fusion gain (best late-fusion temporal micro F1 = 0.4673 vs. 0.6032 for Contact-EC). Retrained simple fusion baselines also remain lower than Contact-EC (Supplementary architecture-baseline table).

3.4. SP-2024 Larger Temporal Evaluation

To verify that the fusion advantage holds at greater temporal scale, we constructed a second temporal holdout from Swiss-Prot entries created after the SP-2023-01 release date. Applying the same closed-set filter—proteins with complete Level-4 EC labels inside the SP-2018 vocabulary—yields 1,226 evaluable proteins from 3,137 total new reviewed entries (39.1% pass rate; the remaining 60.9% carry partial or novel EC labels, a label-shift fraction similar to SP-2023-01).

On SP-2024 ($N = 1{,}226$, three seeds), the ESM-2-only baseline decreases: B1 falls from 0.4508 ± 0.0203 to 0.3892 ± 0.0121 (-6.2 pp). Contact-only B3 is essentially stable (0.4388 ± 0.0166, $+1.4$ pp within seed variance), while Contact-EC flat FC improves to 0.6819 ± 0.0026 ($+5.8$ pp) and Contact-EC-Hier to

0.6253 (+5.6 pp) (Table 4). The gap between fusion and the ESM-2-only baseline widens from 17.3 pp on SP-2023-01 to 29.3 pp on SP-2024.

A MMseqs2 top-hit homology transfer baseline on SP-2024 achieves 0.7080 micro F1, a +12.3 pp *increase* over SP-2023-01 (0.5852), indicating that SP-2024 proteins have sufficient homolog coverage in the SP-2018 training set and are not intrinsically harder retrieval targets. Against this backdrop, HIT-EC falls to 0.4578 on SP-2024 (−38.9 pp), a decrease substantially larger than any Contact-EC change. Contact-EC flat FC is consequently higher than HIT-EC on SP-2024 by +22.4 pp under this mapped evaluation, reversing the −22.3 pp deficit on SP-2023-01. This reversal is consistent with HIT-EC's sequence-only architecture being sensitive to temporal distribution shift as the evaluation horizon extends, while structural topology in Contact-EC may provide complementary evidence not captured by sequence context alone. Bootstrap 95 % confidence intervals (1,000 resamples, single-seed ecbench checkpoint) confirm non-overlapping CIs between Contact-EC [0.6403, 0.6914] and both B1 [0.3520, 0.4117] and B3 [0.3051, 0.3643]; McNemar protein-level tests give $\chi^2 = 327.6$ (CE vs B1) and $\chi^2 = 339.6$ (CE vs B3), both $p < 0.0001$.

To test whether the HIT-EC decrease is explained by vocabulary mismatch, we stratified SP-2024 proteins by whether their true Level-4 EC labels are covered by HIT-EC's 4,255-class vocabulary. Of 1,226 proteins, 1,000 (81.6 %) have *all* true labels inside HIT-EC's vocabulary (all_in_vocab), 29 (2.4 %) are partial, and 197 (16.1 %) have no true label in HIT-EC's vocabulary (all_out_vocab). On the all_in_vocab subset—where HIT-EC can in principle assign the correct EC—HIT-EC achieves only 0.5496 micro F1, a −29.7 pp drop from its SP-2023-01 level of 0.8471. The vocabulary gap accounts for roughly 9.2 pp of the total −38.9 pp decline; the remaining −29.7 pp (75 % of the total drop) remains after restricting to vocabulary-covered proteins. Contact-EC is stable across all strata: 0.6965 on all_in_vocab, 0.5000 on partial, and 0.5396 on all_out_vocab (Table 5).

Table 4 SP-2024 temporal evaluation ($N = 1{,}226$ complete known-EC proteins from 3,137 new Swiss-Prot entries after 2023-01-25). B1, B3, and Contact-EC (flat FC) values are mean ± std across three seeds; Contact-EC-Hier and HIT-EC are single-seed. HIT-EC decreases by −38.9 pp while Contact-EC improves, reversing the ranking seen on SP-2023-01 under this mapped evaluation.

Model	SP-2023-01	SP-2024	Δ
MMseqs2 (top-hit transfer)	0.5852	**0.7080**	+12.3 pp
B1 (ESM-2, flat FC)	0.4508 ± 0.0203	0.3892 ± 0.0121	−6.2 pp
B3 (Contact map only)	0.4244 ± 0.0207	0.4388 ± 0.0166	+1.4 pp
Contact-EC (flat FC)	0.6241 ± 0.0170	$\mathbf{0.6819 \pm 0.0026}$	+5.8 pp
Contact-EC-Hier	0.5690	0.6253	+5.6 pp
HIT-EC (mapped)	0.8471	0.4578	−38.9 pp

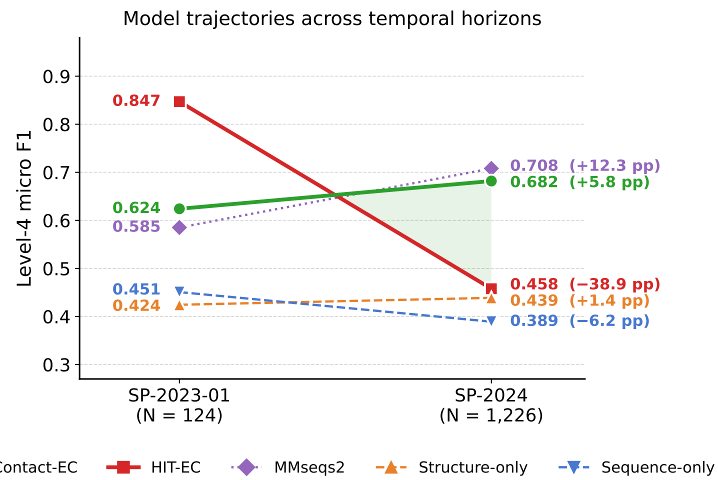

Figure 3 Model trajectories across the two temporal horizons. Each endpoint value and signed change (pp) is annotated in the model's colour. HIT-EC (red) falls −38.9 pp, whereas Contact-EC (green) improves +5.8 pp, producing a +22.4 pp reversal on SP-2024 (shaded region). Thick solid lines: our model and HIT-EC; thin dashed lines: single-modality baselines.

Table 5 HIT-EC vocabulary-stratified evaluation on SP-2024. Proteins are grouped by whether their true Level-4 EC labels fall inside HIT-EC's 4,255-class vocabulary. Even among all_in_vocab proteins (where HIT-EC can in principle succeed), HIT-EC drops −29.7 pp from its SP-2023-01 level (0.8471), indicating that vocabulary mismatch alone does not explain the observed HIT-EC decrease. Stratified Contact-EC values use the canonical seed-42 checkpoint; therefore the aggregate score (0.6661) differs from the three-seed mean in Table 4 (0.6819±0.0026).

Stratum	N	HIT-EC F1	Contact-EC F1
all_in_vocab	1,000 (81.6 %)	0.5496	**0.6965**
partial	29 (2.4 %)	0.1887	**0.5000**
all_out_vocab	197 (16.1 %)	0.0000	**0.5396**
All	1,226	0.4578	**0.6661**

3.5. Foldseek-Disjoint Generalisation

Sequence-disjoint evaluation does not necessarily remove structural fold similarity. On the primary Foldseek/TM-score 0.50 split, all models show a large absolute drop. Across three training seeds, the default Level-4 threshold of 0.5 gives micro F1 of 0.0433 ± 0.0042 for B1, 0.0400 ± 0.0054 for B3, and 0.0872 ± 0.0122 for fusion. Validation-selected thresholds applied once to the held-out Foldseek test split preserve the same ordering, with fusion reaching 0.1685 ± 0.0334 micro F1 versus 0.0607 ± 0.0009 for B1 and 0.0654 ± 0.0016 for B3. A single-seed TM-score sweep at 0.40/0.50/0.60 also preserves the fusion advantage (0.2981/0.0998/0.4403 micro F1), although the 0.50 split has lower label coverage and a different cluster-assignment mode (Supplementary Section S9). In contrast, MMseqs2 top-hit EC transfer is much stronger on the same split (Level-4 micro $\mathrm{F1} = 0.5393$), showing that this audit tests complementary neural generalisation rather than replacement of curated homology transfer.

The split also induces label-coverage shift: 31.5% of positive Level-4 labels in Foldseek test are absent from the corresponding training partition, and 31.6% of test proteins contain at least one unseen Level-4 label. Among seen labels, fusion is strongest for moderately represented classes (0.2484 micro F1 for 6–25 training proteins and 0.3278 for 26–100), indicating that fold-disjoint performance reflects both structural shift and label-frequency shift. A three-seed failure-mode audit further shows that fusion rescues 17.3% of Foldseek-test proteins missed by B1 and 12.5% missed by B3; 11.0% are fusion-only rescues, whereas 41.9% of all-model failures contain an unseen true Level-4 label. When unseen positive labels are excluded from the same Foldseek-disjoint prediction outputs, aggregate seen-label Level-4 micro F1 increases to 0.2159 for fusion versus 0.0863 for B1 and 0.0859 for B3 (Supplementary seen-label aggregate table), showing that the fusion advantage is not solely a consequence of unseen-label composition.

3.6. Topology Controls and Data Recency

To test whether the contact branch uses topology rather than contact-map density or length proxies, we evaluated three models under three perturbed contact-map variants on the temporal test set ($N = 124$): shuffled (symmetric row/column permutation, topology destroyed), diagonal-only (constant band $|i - j| < 12$, no protein-specific long-range contacts), and density-matched random (Table 6).

Table 6 Contact-map perturbation controls (temporal test, Swiss-Prot 2023-01, $N = 124$). B3 drops to zero under all perturbations. Contact-EC flat FC decreases sharply (-54 pp shuffled, -50 pp random density), consistent with reliance on fold topology rather than contact density alone. Contact-EC-Hier shows negligible change: the structure gate routes to the ESM-2 branch when contact topology is destroyed, explaining why flat FC outperforms Hier.

Contact map variant	B3	Contact-EC (flat FC)	Contact-EC-Hier
Original	0.3646	**0.6032**	0.5690
Shuffled (row/col perm.)	0.0000	0.0625	0.5702
Diagonal-only	0.0000	0.5242	0.5776
Density-matched random	0.0000	0.0982	0.5641

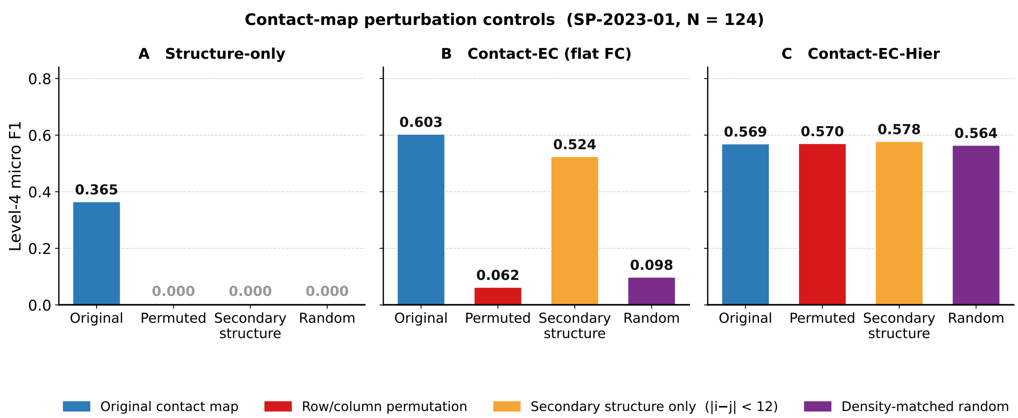

Figure 4 Contact-map perturbation controls (SP-2023-01, N=124), shown separately per model. **(A)** B3 (contact-map only) drops to zero under all three perturbations, consistent with strong sensitivity to contact-map topology and distribution. **(B)** Contact-EC flat FC drops sharply under shuffling (-54.1 pp) and density-matched random (-50.5 pp) but retains most performance under diagonal-only (-7.9 pp), indicating short-range secondary-structure contacts remain informative. **(C)** Contact-EC-Hier is invariant across all conditions (±1.0 pp), revealing that its structure gate routes to the ESM-2 branch when topology is destroyed.

Two findings emerge. First, the flat FC model exhibits massive degradation under shuffling (-54.1 pp) and random density matching (-50.5 pp), confirming that its performance derives from genuine three-dimensional fold topology rather than contact-map density or protein-length proxies. The partial retention under diagonal-only maps (0.5242, -7.9 pp) indicates that short-range contacts ($|i-j| < 12$, encoding secondary structure propensity) contribute substantially, with long-range contacts providing an additional $+7.9$ pp. Second, Contact-EC-Hier is essentially invariant to perturbation (±0.9 pp), revealing that its structure gate has learned to route contact information to zero when topology is disrupted—graceful degradation to ESM-2 performance. This gate behaviour simultaneously explains why Contact-EC flat FC *outperforms* Contact-EC-Hier by 3.4 pp on the temporal test: the hierarchical Transformer head induces cross-level attention dynamics that interfere with contact-branch integration, causing the gate to route sub-optimally even on intact contact maps.

The recent-corpus experiment tests the effect of updating the training corpus and label vocabulary. A recency-cutoff audit shows no accession or exact-sequence overlap between the ExpA training split and the temporal proteins, but the ExpA encoder evaluates only 99 of the 124 complete-known temporal proteins at Level 4. On this matched $N = 99$ intersection, Contact-EC trained on SP-2018 reaches 0.6467 ± 0.0210 micro F1, whereas Contact-EC-ExpA reaches 0.7417 ± 0.0182 across three seeds. Thus the fair-subset recent-corpus gain is $+9.5$ pp, not the naive $+11.8$ pp obtained by comparing different denominators. Sample size, class frequencies, label coverage, and homolog coverage also change, so this is a recent-corpus effect rather than a pure date effect. A separate MMseqs2 nearest-neighbour audit supports this interpretation: on the same $N = 99$ subset, moving from SP-2018 to SP-2022/ExpA increases the median top-hit identity from 0.556 to 0.619, the number of proteins with a ≥ 0.60-identity training neighbour from 38 to 49, and top-hit exact Level-4 EC agreement from 69 to 78. In contrast, end-to-end ESM-2 fine-tuning reaches similar in-distribution validation performance but drops to 0.6703 temporally. This suggests over-adaptation of the PLM backbone under limited supervised data and supports frozen or lightly tuned representations for temporal robustness. The matched $+9.5$ pp recent-corpus gain remains larger than the $+2.8$ pp gain from replacing ESM-2 650M with the 3B backbone, reinforcing that benchmark date, data volume, homolog coverage, and label coverage can matter as much as or more than backbone scale.

4. Discussion

The results support a nuanced interpretation. Contact maps provide a genuine and complementary signal for EC prediction, especially on sequence-disjoint validation, moderately represented fold-disjoint labels, and several broad EC families. On the one-year horizon (SP-2023-01), the strongest temporal score remains HIT-EC (0.8471), and controlled retraining shows that recent-corpus factors explain a substantial part of the gap to Contact-EC (0.6241). This distinction matters for method development: an architecture that improves a fixed-date, closed-set benchmark may still underperform a simpler or sequence-only system trained with a more recent label vocabulary.

Notably, the SP-2024 evaluation ($N = 1{,}226$) complicates this picture in a significant way. HIT-EC, which led by $+22.3$ pp on SP-2023-01, falls to 0.4578 on SP-2024 (-38.9 pp)—a decrease substantially larger than any Contact-EC change. Contact-EC flat FC (0.6819 ± 0.0026) consequently overtakes HIT-EC by +22.4 pp on SP-2024. A MMseqs2 top-hit transfer baseline achieves 0.7080 on SP-2024 (+12.3 pp over SP-2023-01), showing that SP-2024 proteins are not intrinsically harder homology targets—the training corpus provides adequate coverage. This makes HIT-EC's decrease more notable. Vocabulary-stratified analysis (Table 5) indicates that this decrease is not primarily an artefact of HIT-EC's smaller label vocabulary: even among the 1,000 SP-2024 proteins (81.6 %) whose true EC labels are fully inside HIT-EC's 4,255-class vocabulary, HIT-EC achieves only 0.5496—a −29.7 pp drop from SP-2023-01. The vocabulary gap accounts for only ∼9 pp (∼25 %) of the total −38.9 pp decline; the remaining ∼30 pp persists in sequences that HIT-EC's vocabulary nominally covers. Contact-EC maintains strong performance across all vocabulary strata (0.6965, 0.5000, 0.5396 for all_in, partial, and all_out proteins, respectively), consistent with structural topology in Contact-EC providing evidence that may be complementary to sequence-derived features and relatively stable across annual Swiss-Prot release cycles. These findings suggest that structure-aware models may carry a temporal advantage at longer horizons even when sequence-only models lead on shorter ones, and reinforce the case for reporting evaluation horizon alongside headline benchmark scores.

The evaluation also shows why EC benchmark design matters. The same model can look strong on random splits, near-parity on sequence-disjoint validation, and much weaker under temporal shift. Moreover, the full Swiss-Prot 2023-01 temporal set is not a single closed-set benchmark: 344/468 proteins contain partial or novel labels. Future studies should report training cutoff, sequence overlap, fold-level relatedness, label-vocabulary coverage, partial-label frequency, and structure availability alongside headline F1. We recommend reporting both the evaluable closed-set subset and the full temporal label-shift set, because these answer different questions: recognition of known EC classes versus robustness to newly curated or incomplete annotations.

There are limitations. Contact-EC relies on AlphaFold or predicted structures, so performance can degrade when external structures or predicted contact maps come from a shifted distribution. On Price-149 [19], supplementary audit results show that input coverage is available but exact Level-4 specificity drops sharply despite stronger coarse-level EC recovery, indicating an OOD calibration and specificity failure rather than a simple missing-input problem. A calibration audit supports this interpretation: on Price-149, Contact-EC emits nonempty predictions for 51.5% of encoded proteins with median top-1 probability 0.5527, but top-1 hit rate is only 27.9%. Supplementary contact-pair and raw contact-map topology nearest-neighbor audits further show high nearest training-set similarity for nearly all evaluable temporal proteins, explaining why temporal results should not be interpreted as fold-disjoint structural generalization. The explicit Foldseek split addresses this issue, but it also exposes severe label-coverage shift: in the primary 0.50 split, 31.5% of Foldseek-test positive labels are unseen in the corresponding training partition, compared with 15.9% and 10.3% in the 0.40 and 0.60 sweep splits. MMseqs2 top-hit transfer remains substantially stronger than all closed-set neural models, reinforcing that Contact-EC is a decomposition probe rather than a homology replacement. Macro F1 remains low due to the long-tailed Level-4 label space. The temporal test is small; although the main temporal B1/B3/Contact-EC comparison, the matched recent-data audit, and the Foldseek audit now include three-seed repeats, 3B and end-to-end variants are still single-run analyses due to compute cost. Bootstrap and paired tests therefore quantify sample-level uncertainty rather than definitive rankings. Finally, supplementary open-vocabulary audits show that fusion confidence can partially flag unseen-label cases, but closed-set models cannot solve novel EC discovery without retrieval, abstention, or label-expansion mechanisms.

5. Conclusions

This study reframes structure-aware EC prediction as a benchmark decomposition problem. Contact maps alone can match ESM-2 650M on hard sequence-disjoint validation, and sequence–structure fusion substantially improves temporal F1 over either modality alone. Perturbation controls indicate that the structure branch uses fold topology, and Foldseek-disjoint evaluation shows that fusion retains a measurable advantage under fold-level shift even though absolute closed-set Level-4 performance remains low. Data-recency and fine-tuning controls show that annotation cutoff and backbone adaptation strongly affect temporal performance. Critically, an extended SP-2024 evaluation ($N = 1{,}226$) shows that the published sequence-only HIT-EC model decreases by $-38.9\,\mathrm{pp}$ over two-plus years while Contact-EC holds and overtakes it under the mapped evaluation, suggesting that structure-aware representations may provide complementary robustness as the temporal horizon grows. These findings argue for EC prediction benchmarks that explicitly separate representation generalisation from fold shift, label shift, data recency, backbone scale, structure availability, and evaluation horizon.

Declarations

Ethics approval and consent to participate

Not applicable. This study uses publicly available protein sequence, annotation, benchmark, and predicted-structure resources and does not involve human participants, human data, human tissue, animals, or clinical intervention.

Consent for publication

Not applicable.

Availability of data and materials

Swiss-Prot sequences and EC annotations were downloaded from UniProt [20] (`https://www.uniprot.org`). AlphaFold structures were retrieved from the AlphaFold Protein Structure Database (`https://alphafold.ebi.ac.uk`). New-392 and Price-149 benchmark files are available from the CLEAN-Contact repository [13]. EC-Bench training and test data can be obtained from the EC-Bench GitHub repository [21]. Code and reproducibility artifacts for reproducing tables and figures are available at `https://github.com/jamesksh0130/contact-ec`. Project name: Contact-EC. Project home page: `https://github.com/jamesksh0130/contact-ec`. Archived version: GitHub tag `bmc-submission-v1`. Operating system: Linux. Programming language: Python. License: MIT. There are no restrictions on non-academic use beyond the terms of the repository license and the licenses of the underlying third-party datasets and models. At submission, the public GitHub repository is the primary code and reproducibility archive.

Competing interests

The author declares no competing interests.

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Authors' contributions

CRediT author statement: Author: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Visualization, Writing – Original Draft, Writing – Review & Editing. The author has read and approved the final manuscript.

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